Oriented Gaussian Function-Based Box Boundary-Aware Vectors for Oriented Ship Detection in Multiresolution SAR Imagery

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***Abstract*— As an important remote sensing means, synthetic aperture radar (SAR) has many superiorities to other sensors. How to effectively detect and locate ships in SAR images is also a popular field. In previous ship detection research, most algorithms focus on detecting the horizontal bounding box of ship targets, which ignore the rotation angle of each ships. Thus, too much background noise in the horizontal detection results makes them difficult to describe each ship accurately. Inspired by the powerful feature representation ability of convolutional neural networks (CNNs), a novel anchor-free and keypoint-based deep learning method is proposed for oriented ship detection in multiresolution SAR images. Our detector first extracts multilevel features from the input SAR image with a backbone network and feature pyramid network. Next, considering multiscale ships in multiresolution SAR images, we detect different sizes of ships on different levels of feature maps with identical head network structures. In each head network, the classification subnetwork determines each pixel in feature maps as the central pixel of this ship or not, and the regression subnetwork regresses the oriented bounding box for each ship. In the training process, the proposed oriented nonnormalized Gaussian function is used to describe the center point of ship targets, while the nonuniform weighting of the different level loss functions is used to suppress the imbalanced sample distribution. Experimental results on two authoritative SAR-oriented ship detection datasets and two Gaofen-3 images demonstrate the effectiveness and robustness of the proposed methods.**

***Index Terms*— Anchor-free detector, convolutional neural network (CNN), oriented object detection, ship detection, synthetic aperture radar (SAR).**

I. Introduction

SHIP detection has been one of the most heavily researched topics in remote sensing, since it can effectively improve the efficiency of marine transportation and reduce maritime traffic accidents [1]–[3]. As an important remote sensing technology, synthetic aperture radar (SAR) can provide abundant and high-resolution landmark information with the advantages of all-time, all-weather, and long operation distances [4]. Composed of many complex metal components, ship targets usually possess a high radar cross section (RCS), which means that their reflection energy is higher than that of other targets or backgrounds in SAR images [5], [6]. These salient advantages make SAR ship detection a powerful means for maritime surveillance.

Although realizing ship target detection using SAR systems has obvious advantages, many difficulties limit the ship detection performance. The complex sea clutter caused by severe waves highly affects the quality and visuality of SAR images [6], [7], further degenerating the reliability of usual ship detection methods, as shown in Fig. 1(a). In addition to sea clutter, the complicated land background, such as buildings and harbors in Fig. 1(b), reflects strong scattering energy that could lead to high false alarms for inshore ship detection [8]. Most importantly, ship targets in SAR images often appear with arbitrary orientations depending on the perspective of the SAR system platforms [7], [9]. The commonly used horizontal object detection method introduces too much background clutter into the detection result, as shown in Fig. 1(c). To overcome this problem, the oriented angle of ships should be determined, and the oriented bounding box should be detected, as shown in Fig. 1(d). Thus, the accurate location of oriented ships is a challenging task that is urgently solved.

A. Current Research

According to the difference in utilized features, ship detection methods can be roughly categorized into statistical model-based methods and deep learning (DL)-based methods. Traditional statistical model-based methods, spectral residual [10], constant false alarm rates (CFARs) [5], [6], [11], and the improved algorithms derived from them have been widely used. The grayscale feature-based CFAR methods [5] fully consider the reflection energy difference between ship targets and other background regions. Inspired by this CFAR method, many typical CFAR detectors have been proposed considering different intensity distributions in sliding windows, e.g., cell average CFAR (CA-CFAR) [11] and order statistics-based CFAR (OS-CFAR) [5]. Meanwhile, CFAR [1], [12], [13] methods have been incorporated with diverse modules to

Manuscript received March 5, 2021; revised May 11, 2021, June 1, 2021, and June 28, 2021; accepted July 3, 2021. Date of publication July 16, 2021; date of current version January 14, 2022. This work was supported in part by the National Key Research and Development Program of China, in part by the Joint Funds of the National Natural Science Foundation of China under Grant U19B2018, in part by the National Science Fund for Distinguished Young Scholars under Grant 61825105, and in part by the Seed Funding Project of Multisensor Intelligent Detection and Recognition Technologies Research and Development Center of CASC under Grant ZZJJ201902-01. *(Corresponding author: Mengdao Xing.)*

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Digital Object Identifier 10.1109/TGRS.2021.3095386

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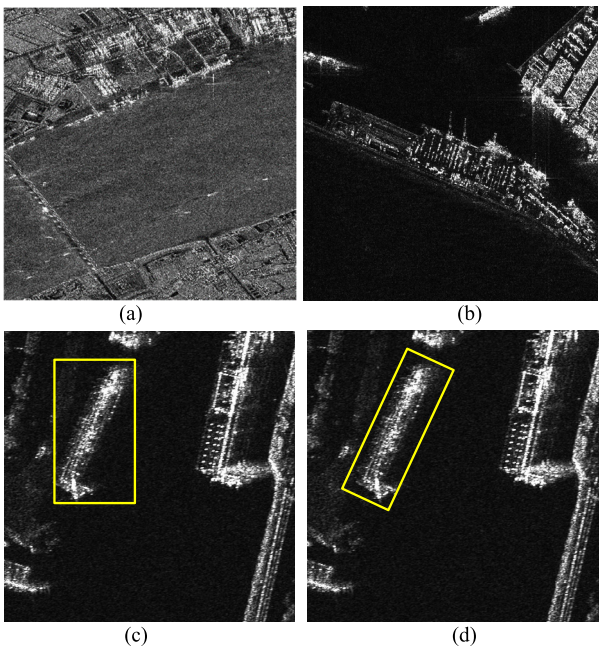


Fig. 1. Few conditions limit ship detection performance. (a) Complex sea clutter in SAR image. (b) Complicated land background in SAR image. (c) Background clutter is introduced into the horizontal detection result. (d) Oriented ship detection can remove the clutter in the horizontal detection result.

improve detection precision for specific needs. In addition to these CFAR-based methods, image feature-based methods such as edge-enhanced maximally stable extremal regions (EMSERs) [14] and complex coherence among subapertures are also used in SAR ship detection. These statistical-based or handcrafted-based methods perform well under the homogeneous sea or weak clutter. However, regarding high-resolution SAR images, complex backgrounds such as isolated islands, coastlines, and strong sea waves will result in high false alarms [7], [8], [15].

Recently, DL [16]–[19] has attracted much research attention due to its powerful feature representation ability. In particular, the convolutional neural network (CNN) shows high robustness and efficiency in optical natural image processing [19]–[21]. Meanwhile, CNN-based detection methods have also gained tremendous development. According to the difference between head structures, the CNN-based detection method can be categorized into two-stage detection methods and one-stage detection methods [9], [22]. For the two-stage detection methods, Faster R-CNN and its variants [9], [13], [23] have shown comparative advantages in SAR ship detection. Kang *et al*. [13] creatively combine Faster R-CNN with CFAR by taking the candidate region proposals generated by Faster R-CNN as the guard windows of CFAR. Li *et al*. [9] proposed an improved Faster R-CNN method and provided a publicly available SAR Ship Detection Dataset (SSDD). Cui *et al*. [24] proposed a dense pyramid network based on a pyramid structure. Sun *et al.* [19] proposed a part-based CNN for complex composite object detection. Meanwhile, Cui *et al.* [25] introduced the spatial shuffle-group enhance attention module into CenterNet for ship detection in large-scale SAR images. For the one-stage methods, the representative methods are You Only Look Once (YOLO), Single Shot MultiBox Detector (SSD), and RetinaNet [26]. Inspired by the YOLO series, Zhang and Zhang [27] proposed the grid CNN for real-time detection. Two-stage ship methods usually perform high detection performance, while one-stage methods have a fast running speed [7].

B. Problem and Motivation

The above canonical DL-based methods have achieved significant success. However, some disadvantages limit their applications in SAR ship detection.

1) Although some CNN-based ship detection methods consider the difference of ship sizes and propose corresponding strategies [24], [26], e.g., introducing the FPN into a feature extraction network, how to effectively detect multiscale ship objects is still a difficult problem. Meanwhile, the number imbalance between different scale samples also increases the difficulty of ship detection [7], [9], [28].
2) The most commonly used anchor-based methods generate too many anchors covering the image densely to maximize the recall rate, which is inconsistent with ship detection in SAR images because the ship target in SAR images is usually sparsely distributed, and most regions in SAR images belong to the background region [9], [13]. Thus, the anchor-based methods aggravate the sample imbalance and lead to a heavy computational burden, which makes them not the best choice for SAR ship detection [25].
3) Canonical ship detection methods usually locate each ship object with the minimum horizontal bounding box [15], [26], [29]. However, when the ship object is not horizontal or vertical, too much background clutter will be introduced into the horizontal bounding box, which not only decreases the detection accuracy but also affects the postprocessing steps [2], [30]. Thus, it is necessary to detect oriented ship objects with oriented bounding boxes.

C. Contribution

In this article, we propose an anchor-free SAR ship detection method. As shown in Fig. 2, this network first extracts multilevel features from the input SAR image with a backbone network, and then an FPN hierarchically combines the low-dimensional but high-resolution features and high-dimensional but low-resolution features with skip connections. After that, we obtain three groups of feature maps, $P2$, $P3$, and $P4$, and their corresponding total strides are 4, 8, and 16, respectively. Following FPN, we detect different sizes of ship objects on different levels of feature maps with identical head network structures. In a head network, the classification subnetwork determines each pixel in feature maps as or not the center pixel of the ship object and the regression subnetwork regresses the oriented bounding box for each object. Thus, the proposed network is an anchor-based network that is free of devising complicated anchors. The main contributions of this article are as follows.

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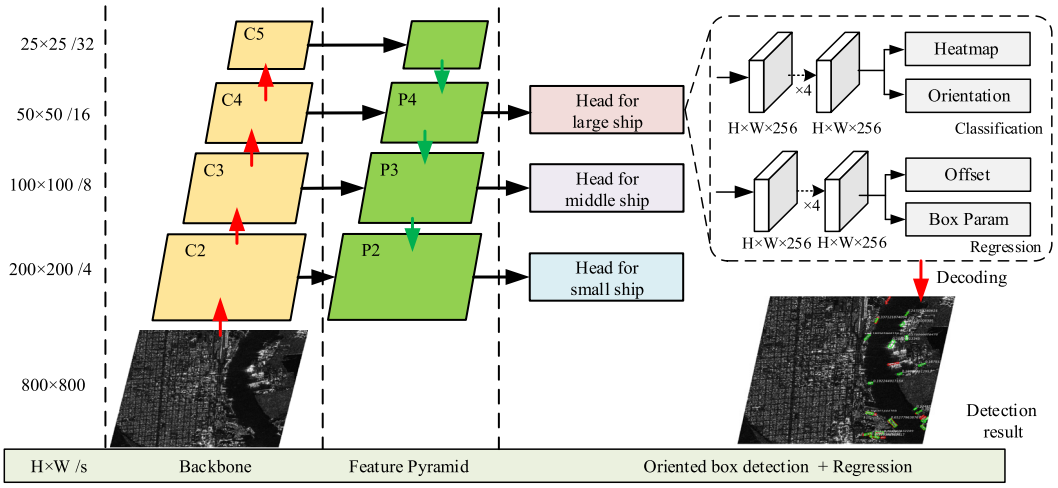


Fig. 2. Network structure of the proposed method.

1) An end-to-end network structure is proposed for oriented SAR ship detection. The hierarchical detection strategy in this structure detecting different sizes of ship objects on different FPN levels of feature maps can improve the accuracy of multiscale ship target detection against complex backgrounds.
2) Traditional keypoint-based anchor-free methods try to reduce the penalty given to the negative locations with an unnormalized 2-D Gaussian focusing on the center point of objects (positive location), which ignore the oriented angle of ship targets and lead to an unequal penalty of object region. To suppress this problem, we propose an oriented nonnormalized Gaussian function, whose variance and covariance are related to the oriented angle and scale of ship objects. This function can decrease the sensitivity to the position drift of the center point and improve the recall rate of ship detection.
3) Keypoint-based box boundary-aware vectors (BBAVs) are introduced to our head networks to describe oriented ship objects with oriented bounding boxes. These vectors distributed in the four quadrants of a Cartesian coordinated system incorporating FPN can help us to accurately detect multiscale oriented ships while maintaining a fast detection speed. Moreover, the nonuniform weighting factor of the loss function from different head networks helps to suppress the imbalance between different sizes of ship objects.

The rest of this article is organized as follows. Section II introduces the related works. Section III illustrates our proposed method. Section IV introduces the oriented SSDD and describes the corresponding experimental details. Section V presents the conclusions.

II. Related Work

Our method is related to two state-of-the-art CNN-based detection methods: anchor-free detectors and oriented object detection. To stress our contributions to these powerful methods, they are introduced in this section.

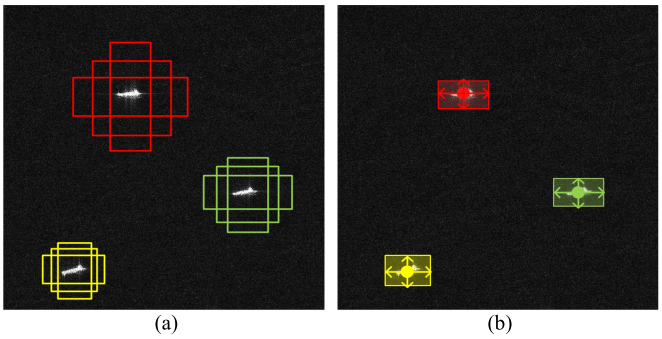


Fig. 3. (a) Anchor-based detectors generate many overlapping anchors. (b) Anchor-free detectors only detect the keypoints of objects.

A. Anchor-Free Detectors

The high resolution of SAR images usually leads to ship objects being sparsely distributed in these images, which means that the anchor-based detectors will generate too many false alarm anchors and lead to a heavy computational burden [as shown in Fig. 3(a)] [23], [26]. To overcome the above issues, anchor-free detectors are used in this article to detect ship objects from SAR images. There are two main groups of detectors in anchor-free object detection. The first group of detectors, belonging to the top-down approach, directly predicts the bounding boxes and category (including background) at each location in the final feature maps. The canonical top-down detectors are FSAF [31], FoveaBox [32], FCOS [20], and so on. In particular, their good portability and powerful performance make FCOS a popular top-down detector.

The second group of anchor-free detectors is the keypoint-based bottom-up method, which first detects the keypoints of objects, such as corners, center, and extreme points, and then groups the detected keypoints to yield whole-object detections, as shown in Fig. 3(b). The canonical bottom-up anchor-free detectors are CornerNet [22], CenterNet [25], [33], ExtremeNet [34], and so on. CornerNet detects an object similar to a pair of keypoints: the top-left corner and bottom-right corner of the bounding box. For the ground truth positive

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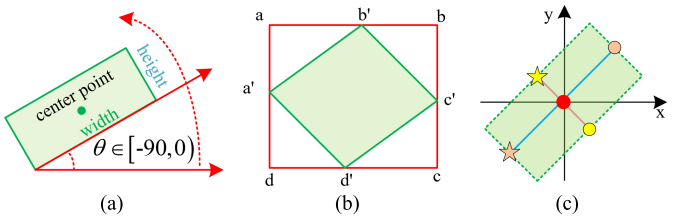


Fig. 4. Several definitions of oriented bounding boxes. (a) Five-parameters method with 90° angular range. (b) Eight-parameter ordered quadrilateral representation. (c) Eight-parameter middle lines representation.

location of each corner, CornerNet considers the tolerance of the bounding box and reduces the penalty given to the negative locations within a radius of the positive location and an unnormalized 2-D Gaussian. Inspired by CornerNet, CenterNet [25] tries to detect the center point of the bounding box and regresses the size, orientation, and posture with the extracted center points.

Inspired by FCOS and CenterNet, we detect the center point of different sizes of ship objects on different levels of feature maps. Our contributions and detailed steps are described in Section III.

B. Oriented Object Detection

Currently, oriented object detectors are mainly derived from anchor-based or anchor-free detection methods. Among these detectors, multistage rotation detection methods such as ROI-Transformer [30], SCRDet [35], and R^3Det [36] are still dominant due to their excellent robustness. The effective definition of oriented bounding boxes plays a very important role in oriented object detection. Several bounding box definitions are shown in Fig. 4. The five-parameter model with (x, y, w, h, θ) representation [37] shown in Fig. 4(a) is the most widely used definition, where (x, y) is the position of the center point, (w, h) is the width and height of the bounding box, and θ denotes the acute angle to the x-axis. The eight-parameter ordered quadrilateral representation shown in Fig. 4(b) directly regresses the four corners of the object, which is used in the Gliding vertex [38]. The oriented objects detection network detects oriented objects by predicting a pair of middle lines inside each target [39], as shown in Fig. 4(c). To overcome the discontinuous boundaries problem of the five-parameter method and eight-parameter method [40], Yi *et al.* [41] proposed a BBAV method that first detects the center point of the object and then regresses four vectors in four quadrants of a Cartesian coordinate system. Although the BBAV method has a better performance than other traditional methods, it ignores the size difference of training samples. Meanwhile, the center point ground truth of objects in BBAV is still generalized by the usual unnormalized 2-D Gaussian, which ignores the center point drift along the oriented angle of oriented objects.

In this article, we introduce the BBAV representation into our FPN-based anchor-free detector to detect oriented ship objects with different sizes in different-resolution SAR images. Moreover, considering that the position sensitivity of the center point can highly affect the detection performance, we propose an oriented unnormalized 2-D Gaussian function to generate a drift region for each center point, which can effectively improve the detection performance.

TABLE I
NOTATION

x, y	Center point of an oriented bounding box
w, h	Width and height of an oriented bounding box
θ	Rotation angle of an oriented bounding box
w_e, h_e	Width, height of minimum bounding rectangle on feature map
c_x, c_y	Center point of bounding box mapping on feature maps
p_x, p_y	Position index on feature maps
l	Loss function in the training process
t, **r**, **b**, **l**	Top, right, bottom, and left box vectors of bounding box
d	Box parameter vector of bounding box
o	Offset vector for the groundtruth of an object
P	Feature map tensor
B	Box parameter tensor on feature maps
A	Orientation parameter tensor on feature maps
O	Offset parameter tensor on feature maps
W	Weighing parameter tensor

III. Methodology

This article adopts the notation summarized in Table I. A detailed description of these symbols is shown below.

A. Feature Extraction and Multilevel Prediction With FPN

It is noticeable that we only use the magnitude fraction of SAR images in our detector. For a CNN-based detector, the choice of backbone network can highly affect the final detection performance [18], [19], [21]. The canonical feature networks are VGGNet [19], [42] and ResNet [43]. Among these networks, ResNet-50, one version of ResNet, has been widely used for its powerful feature representation ability and excellent task adaptability [30], [33], e.g., object classification [18], target detection, and semantic segmentation [44]. Combined with multiple convolutional modules, ResNet-50 downsamples the input image to different-resolution feature maps with pooling layers. With strides of 4, 8, 16, and 32, the corresponding feature maps are represented by $C2$, $C3$, $C4$, and $C5$, respectively. We also use ResNet-50 to extract features from input SAR images. Given the input image in Fig. 5(a), the feature maps extracted by our well-trained detection model of $C2$ and $C3$ are shown in Fig. 5(b) and (c), respectively. One can see that the former layers, such as Fig. 5(b), mainly extract low-dimensional contour and edge features, while the latter layers, such as $C4$ and $C5$, mainly extract high-dimensional keypoint features and semantic features. Although the feature maps in $C5$ are the most widely used in object detection due to their powerful feature representability, the low resolution of $C5$ limits the location accuracy of the bounding box, especially for ship detection in SAR images.

Noticing that the low-level features with abundant location information are useful to object localization, while high-level features with abundant semantic information can improve the accuracy of classification [19], [21], the FPN structure [45] is introduced and redesigned in our feature extraction network to extract more discriminative features, and these networks are shown in Fig. 6. We can obtain three groups of feature

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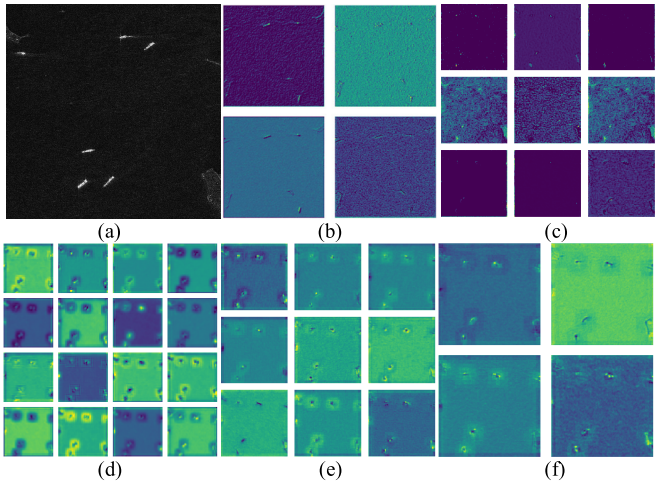


Fig. 5. Input SAR image and corresponding extracted feature maps. (a) Input image with size 800 × 800 pixels. (b) Feature maps of $C2$ with size 200 × 200 pixels. (c) Feature maps of $C3$ with size 100 × 100 pixels. (d) Feature maps of $P4$ with size 50 × 50 pixels. (e) Feature maps of $P3$ with size 100 × 100 pixels. (f) Feature maps of $P2$ with size 200 × 200 pixels.

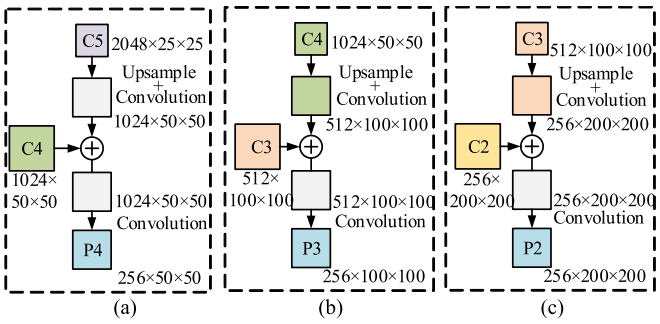


Fig. 6. FPN structure in our detector. The feature maps of (a) $P2$, (b) $P3$, and (c) $P4$ are attained. The 2048 × 25 × 25 for $C5$ means that there are 2048 feature maps with 25 × 25 pixels in $C5$. This representation is also applicable to $C4$, $C5$, $P2$, $P3$, and $P4$. The symbol "⊕" means adding operation.

maps: $P4$, $P3$, and $P2$ by FPN. Taking the generation of $P4$ as an example, the $C5$ feature maps are first upsampled by nearest-neighbor interpolation and convolved by 1024 3 × 3 convolution kernels; then, the convolved feature maps are added by $C4$ maps. Finally, the 3 × 3 convolution kernels reduce the number of feature maps from 1024 to 256. Thus, we obtain 256 feature maps with a size of 50 × 50 pixels for $P4$. Similarly, $P3$ and $P2$ can be generated by the same strategy. A few feature maps in $P4$, $P3$, and $P2$ are shown in Fig. 5(d)–(f), respectively. One can see that the features of ship objects in $P2$, $P3$, and $P4$ are more remarkable than those in $C2$, $C3$, and $C4$, respectively. Thus, the FPN helps to obtain more powerful and accurate features.

According to the pixel area in SAR images, ship targets can be divided into three classes: small targets with a minimum bounding rectangle (MBR) area lower than 1024, middle targets with an MBR area lower than 4096 and higher than 1024, and large targets with an area higher than 4096. The different scales of targets usually express different features in each subnetwork of FPN. To suppress this problem, different sizes

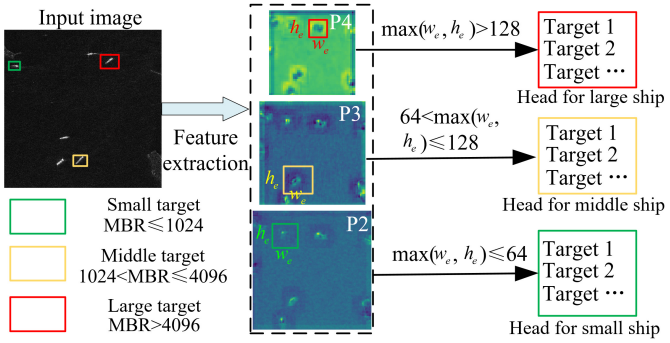


Fig. 7. Multilevel prediction strategy in our detector. Different sizes of ship objects are detected on different levels of feature maps in our detector.

of ship objects are detected on different levels of feature maps in our detector, as shown in Fig. 7. More specifically, we first compute the regression targets w_e and h_e for each location on each feature map level where w_e and h_e are the width and height of the MBR on the feature map, respectively, as shown in Fig. 10(c). Then, if a location follows $\max(w_e, h_e) > m_{i-1}$ and $\max(w_e, h_e) \leq m_i$, it is set to a positive sample that should be optimized by the feature maps with level i. m_i is the maximum distance that the feature level i needs to regress. In our detector, m_1, m_2, m_3, and m_4 for feature maps $P2$, $P3$, and $P4$ are set as 0, 64, 128, and ∞, respectively. With this setting, ship targets with different sizes are responded by different feature maps.

B. Heatmap Based on Oriented Nonnormalized Gaussian

In this article, heatmaps are utilized to locate the center points of each arbitrary ship object in the input SAR images, like in CenterNet [33]. Specifically, the heatmap $\mathbf{P} \in \mathbb{R}^{k\times h\times w}$ generated by the head network has k channels for each group of feature maps with a size of $w \times h$. Since there is only one category target in the ship detection task, k is 1 here, and this channel is passed through a sigmoid function.

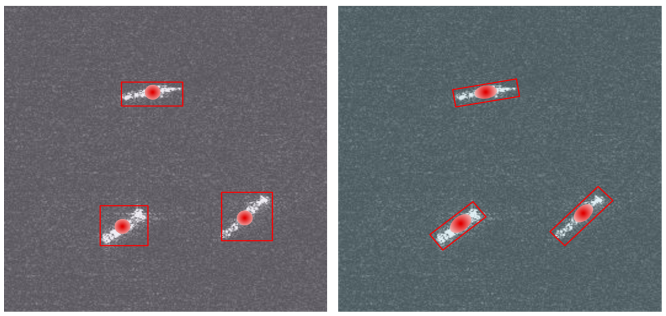
Assuming (c_x, c_y) is the center point of an oriented bounding box, traditional oriented object detection methods [22], [33] use the 2-D nonnormalized Gaussian function $f(c_x, c_y, \sigma)$ around each (c_x, c_y) to represent the heatmap $\mathbf{P} \in \mathbb{R}^{1\times h\times w}$

$$f\left(c_x, c_y, \sigma\right) = \exp\left(-\frac{(p_x - c_x)^2 + \left(p_y - c_y\right)^2}{2\sigma^2}\right) \quad (1)$$

where (p_x, p_y) indexes the position on $\mathbf{P}$. σ is a box size-adaptive standard deviation decided by the IOU between the heatmap and ground truth [22].

With this Gaussian function, if the detected center points are close to their respective ground truth locations, these locations can produce keypoints that sufficiently help to locate ground truth annotation [22], [33], as shown in Fig. 8(a). Thus, $f(\cdot)$ can decrease the sensitivity to keypoint position offset and improve the performance for HBB detection. However, regarding OBB detection, $f(\cdot)$ ignores the oriented angle of ship objects, which leads to an undesirable penalty for negative locations. To suppress this problem, the Gaussian function

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(a) (b)

Fig. 8. Ground truth on feature maps by different Gaussian functions. (a) Ground truth by 2-D nonnormalized Gaussian $f(\cdot)$. (b) Ground truth by our oriented 2-D nonnormalized Gaussian $g(\cdot)$ distribution.

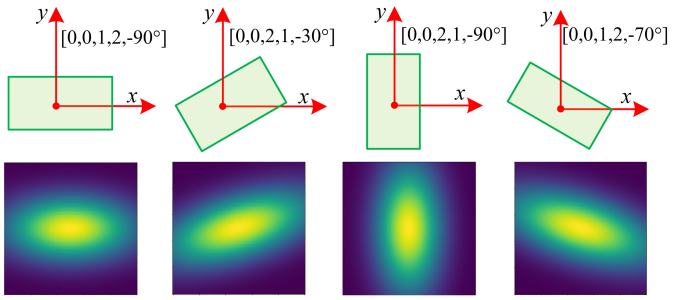


Fig. 9. Generated ground truth by the proposed oriented unnormalized Gaussian function for objects with different oriented angles.

should be oriented and related to the oriented angle of each ship object, as shown in Fig. 8(b). Inspired by this point, our oriented unnormalized Gaussian is proposed as

$$g\left(c_x, c_y, \sigma_x, \sigma_y\right) = \exp\left(-\left(\frac{(p_x - c_x)^2}{2\sigma_x^2} - \frac{\rho \cdot (p_x - c_x) \cdot \left(p_y - c_y\right)}{2\sigma_x \sigma_y} + \frac{\left(p_y - c_y\right)^2}{2\sigma_y^2}\right)\right) \tag{2}$$

where σ_x and σ_y are the x and y spreads of the blob, respectively. ρ is the orientation control coefficient falling into $\{-1, 0, 1\}$. Defining the bounding box as the (x, y, w, h, θ) representation in Fig. 4(a), σ_x and σ_y are decided as follows:

$$\begin{cases} \sigma_x = \sigma, \sigma_y = \sigma \cdot \max\left(\frac{w}{h}, \frac{h}{w}\right), & \text{if } (w-h) \cdot \left(|\theta| - \frac{\pi}{4}\right) > 0 \\ \sigma_x = \sigma \cdot \max\left(\frac{w}{h}, \frac{h}{w}\right), \sigma_y = \sigma, & \text{if } (w-h) \cdot \left(|\theta| - \frac{\pi}{4}\right) < 0 \\ \sigma_x = \sigma_y = \sigma, & \text{otherwise.} \end{cases} \tag{3}$$

In (1), ρ is 1 when $w < h$, and ρ is -1 when $w > h$. In particular, if θ is $-90°$, $\rho = 0$ means that this object belongs to HBB detection. The objects and corresponding ground truth generated by the proposed oriented unnormalized Gaussian function $g(\cdot)$ are shown in Fig. 9.

In Fig. 9, one can see that the proposed function $g(\cdot)$ can generate a wide range that focuses on the center points of each object. Through this function, the generated ground truth is related to the scale and orientation of the ship objects. When training these heatmaps, we decrease the penalty for the points inside the Gaussian bumps in (3) and use the focal loss in (4) to suppress the imbalance between positive and negative samples for training the heatmap

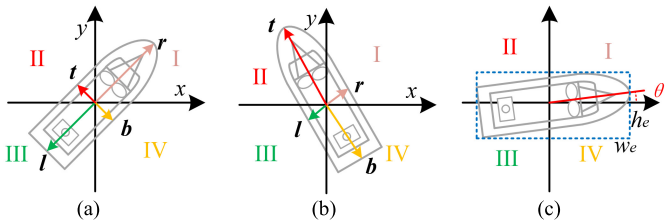


Fig. 10. OBB description for ship objects by BBAV. **t**, **l**, **r**, and **b** in (a) and (b) are the top, right, bottom, and left BBAVs, respectively, which are defined in four quadrants of the Cartesian coordinate system. For nearly horizontal or vertical ship objects, as in (c), the BBAV only regresses the width and height of the external horizontal box.

$$l_h = -\frac{1}{N}\sum_i \begin{cases} (1-p_i)^{\alpha}\log(p_i), & \text{if } \hat{p}_i = 1 \\ (1-\hat{p}_i)^{\beta} p_i^{\alpha}\log(1-p_i), & \text{otherwise} \end{cases} \tag{4}$$

where β is the balance factor used to suppress the imbalance between positive samples and negative samples and α is the weighting factor, which makes this function focus on difficult samples [41].

C. BBAVs and Nonuniform Weighting of Loss Function

After detecting the center point of the ship object, BBAVs are used to describe the OBB of ship objects [41]. BBAV for ships with different angles is shown in Fig. 10.

Instead of the two vectors in Fig. 4(c), the four vectors top **t**, right **r**, bottom **b**, and left **l** in BBAVectors are distributed in four quadrants of the Cartesian coordinate system, which can capture more mutual and salient information for irregular ship images [41]. According to these representation methods, the box parameters are defined as

$$\mathbf{d} = \left[t_x, t_y, r_x, r_y, b_x, b_y, w_e, w_h\right] \tag{5}$$

where t_x and t_y are the coordinates of vector **t.** w_e and w_h are the external horizontal box scales of an OBB. Thus, we construct a box parameter map $\mathbf{B} \in \mathbb{R}^{10\times w\times h}$ for each feature map. The smooth L_1 loss is used to regress the box parameters

$$l_b = \frac{1}{N}\sum_{k=1}^{N} \text{Smooth}_{L_1}\left(\mathbf{d}_k - \hat{\mathbf{d}}_k\right). \tag{6}$$

Due to the pooling layers in the feature extraction network, a location (x, y) in the input SAR image is mapped to the location $\left(\lfloor x/s \rfloor, \lfloor y/s \rfloor\right)$ in the heatmap with a stride of s. To improve the precision when remapping the locations from the heatmap to the input images, an offset map $\mathbf{O} \in \mathbb{R}^{2\times h\times w}$ is predicted as follows:

$$\mathbf{o}_k = \left(\frac{x_k}{s_i} - \left\lfloor \frac{x_k}{s_i} \right\rfloor, \frac{y_k}{s_i} - \left\lfloor \frac{y_k}{s_i} \right\rfloor\right) \tag{7}$$

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where $\mathbf{o}_k$ is the offset for the ground truth of object k. s_i is the total stride of heatmap i. Training this offset map $\mathbf{O}$ is optimized with the smooth L1 loss

$$l_{\text{off}} = \frac{1}{N}\sum_{k=1}^{N} \text{smooth}_{L_1}(\mathbf{o}_k, \hat{\mathbf{o}}_k). \tag{8}$$

By combining the center point and corresponding box parameters, we can locate ship objects and define their bounding boxes. However, when the ships are nearly vertical or horizontal, the boundary problem still occurs, leading to an unstable training process. As in [41], we use a discrimination factor $\hat{\alpha}$ to classify the OBB into a rotation bounding box (RBB) and HBB

$$\hat{\alpha} = \begin{cases} 1 \text{ (RBB)}, & \text{if IOU(OBB, RBB)} < 0.95 \\ 0 \text{ (OBB)}, & \text{otherwise.} \end{cases} \tag{9}$$

Based on the definition $\hat{\alpha}$, the orientation map $\mathbf{A} \in \mathbb{R}^{1\times w\times h}$ is constructed. Meanwhile, the binary cross-entropy loss is used to optimize the orientation map

$$l_\alpha = -\frac{1}{N}\sum_{i=1}^{N} (\hat{\alpha}_i \log(\alpha_i) + (1-\hat{\alpha}_i)\log(1-\alpha_i)). \tag{10}$$

It is noticeable that if $\hat{\alpha}$ is 1 for an object, all ten parameters in box parameters $\mathbf{d}$ should be optimized, and if $\hat{\alpha}$ is 0 for an object, only w_e and w_h in box parameters $\mathbf{d}$ are optimized.

According to the head structure, the loss function for each level feature map i is

$$l^i_{\text{all}} = l^i_h + l^i_{\text{off}} + l^i_b + l^i_\alpha. \tag{11}$$

Traditionally, l^i_{all} for each level feature map is equally summarized to obtain the final loss [20]. However, when the number of training samples assigned to each level varies greatly, uniform weighting leads to unsatisfactory performance. Specifically, when there are too many small samples in the dataset, the head network in the detector focuses on the detection of these small samples and ignores the importance of middle and large samples. Under this circumstance, the learning efficiency for training samples is not adequate [16]. To tackle this problem, nonuniform weighting is proposed as follows:

$$l_{\text{total}} = \alpha_1 l^{P_2}_{\text{all}} + \alpha_2 l^{P_3}_{\text{all}} + \alpha_3 l^{P_4}_{\text{all}} \tag{12}$$

where α_1, α_2, and α_3 are the weighting factors for $P2$, $P3$, and $P4$, respectively. Specially $\alpha_1 = \alpha_2 = \alpha_3 = 1$ corresponds to the uniform weighting. According to (12), the gradient update values of the weighting parameters are as follows:

$$\Delta\mathbf{W}_{P_2} = \alpha_1 \frac{\partial l^{P_2}_{\text{all}}}{\partial \mathbf{W}_{P_2}}, \quad \Delta\mathbf{W}_{P_3} = \alpha_2 \frac{\partial l^{P_3}_{\text{all}}}{\partial \mathbf{W}_{P_3}}, \quad \Delta\mathbf{W}_{P_4} = \alpha_4 \frac{\partial l^{P_4}_{\text{all}}}{\partial \mathbf{W}_{P_4}}. \tag{13}$$

Thus, we can adjust the gradient update value for each level of feature maps by controlling the weighting factors α_1, α_2, and α_3. For instance, if the number of large samples is smaller than that of middle samples and small samples, α_3 should be set higher than 1. Using the loss function in (12), we can obtain a well-trained detection model for oriented ship detection.

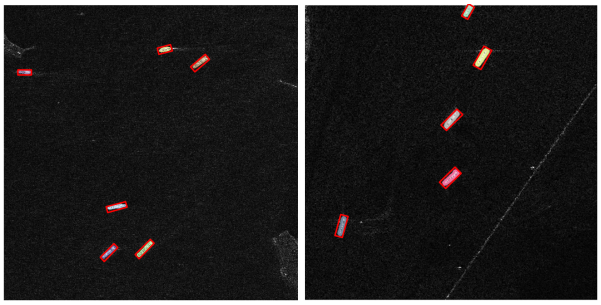


Fig. 11. Two images from HRSID. The original HRSID annotates the polygon of each ship, while we use the MBR corresponding to the polygon annotations to detect oriented ship targets.

IV. Experimental Results

To evaluate the effect of the proposed method, experiments are implemented on the public SSDD. First, the implementations of the dataset, evaluation criteria, and detailed experimental settings are introduced. Next, the role of each module in our detector is verified. Then, traditional ship detection methods are compared to verify the effectiveness of our method. At last, the robustness of the proposed method in SSDD+ and large-scene images is verified.

A. Oriented Ship Detection Dataset and Evaluation Criteria

For DL-based SAR ship detection methods, the detection dataset has played key roles, as it can heavily affect the accuracy and robustness of ship detection methods. The commonly used SAR ship datasets are the OpenSAR ship [28], the SAR-Ship dataset [46], the SSDD [9], and the AIR-SARShip-1.0 dataset [47]. These excellent ship datasets have greatly promoted the development of SAR ship classification methods and ship detection methods. However, some disadvantages limit their applications in CNN-based detectors, e.g., the small image size of OpenSAR ship and SAR-Ship-Dataset leads to that they contain less land and harbor information, and the limited ship objects of SSDD always lead to too high performance and lower generalization ability. Especially, the SAR images in AIR-SARShip-1.0 cover complicated backgrounds, such as harbors, islands, and reefs. The AIR-SARShip data have promoted the development of the DL-based ship detection method. However, the horizontal annotation boxes in AIR-SARShip do not consider the oriented angle of ship targets and limit its application in oriented ship detection. To suppress the above problems, the high-resolution SAR image dataset (HRSID) [7] is mainly used in this article for training and testing our detector. Cropping 136 panoramic high-resolution SAR images with range resolution from 1 to 5 m to 5604 slice images with 800 × 800 pixels, HRSID covers different polarization, imaging modes, and imaging conditions and contains 16 951 ships. Two images from HRSID are shown in Fig. 11.

Since the original HRSID is designed for CNN-based instance segmentation and horizontal ship detection, we take the MBR of the annotated polygon for each ship object as the ground truth of oriented ship detection, as shown in Fig. 11. The area distribution and aspect ratio (length divided by width)

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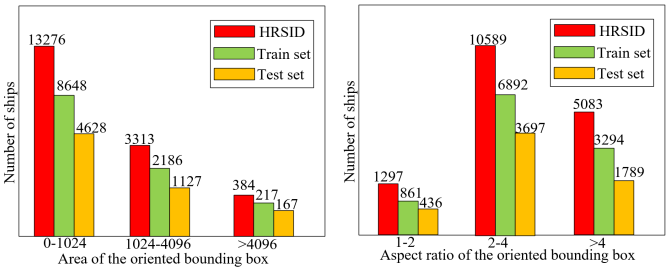


Fig. 12. Distribution of the training set and test set in HRSID, including the aspect ratio and area of the oriented bounding boxes.

distribution of the oriented bounding boxes in the HRSID are shown in Fig. 12. Notably, 3642 HRSID images are used as a training set, while the other images are used as the testing set. Both the area and the aspect ratio face the problem of imbalance distribution.

The same mean average precision (mAP) calculation as for PASCAL VOCs is usually used for the evaluation metrics of OBB detection. Given the IOU of 0.5, mAP is defined as

$$\text{mAP} = \int_0^1 P(R)dR \tag{14}$$

where R indicates the recall rate and P suggests the precision rate. Although this mAP considers the detection performance caused by different confidence levels, the target size and location accuracy of the bounding box are ignored. To better validate the performance of the proposed method, AP evaluation similar to the COCO dataset is introduced in this article. Specifically, AP_{50} and AP_{75} are computed according to IOUs of 0.5 and 0.75, respectively. Meanwhile, the AP is attained by averaging over six IOU thresholds (from 0.5 to 0.75) and tends to reward models that are better at a precise location. In addition to AP, AP_{50} and AP_{75}, we also compute the AP for small, middle, and large targets, named, AP_S, AP_M, and AP_L, respectively. It is noticeable that IOU here means the intersection-over union between detected oriented ship objects and corresponding ground truth.

B. Experimental Settings and Ablation Study

The PyTorch framework is used in this article to implement the proposed method. The size of the input image is set at 800 × 800 pixels. The weights of the ResNet50 backbone are initialized with the pretrained model on the ImageNet dataset [48], while the other weights are initialized under the default settings. The training images in HRSID are augmented by random flipping and random cropping within the scale range [0.9, 1.1]. The Adam algorithm with an initial learning rate of 0.000125 is used to optimize the loss function l_{total}. The number of epochs is 80, while the batch size is 8 on 2 NVIDIA TITAN XP.

Compared with the original BBAV method [41], there are three key modules of the proposed methods: the combination of FPN and multi prediction for detecting multiscale ships, the oriented nonnormalized Gaussian function for locating the center point of ship targets, and the nonuniform weighting of different level loss functions for suppressing the imbalanced

TABLE II
EXPERIMENTAL RESULTS OF EACH COMPONENT OF THE PROPOSED METHOD (%)

		Baseline	Baseline+different settings		
FPN with multilevel prediction			✓	✓	✓
Oriented Gaussian function				✓	✓
Nonuniform weighting					✓
Accuracy	AP	59.45	60.07	60.43	**60.71**
	AP_{50}	77.29	78.22	78.55	**78.60**
	AP_{75}	35.76	36.40	36.92	**37.22**
	AP_S	51.34	54.03	55.34	**56.86**
	AP_M	67.66	63.57	65.47	**67.63**
	AP_L	30.09	**33.46**	30.85	31.12
Efficiency	Params (M)	53.42	32.51	32.51	32.51
	Multi-adds(G)	305.19	586.40	586.40	586.40

sample distribution between different scale targets. To validate the effectiveness of these modules, the ablation study is analyzed in this section. Taking BBAV as the baseline, the results obtained by gradually adding the above three components are shown in Table II.

1) *FPN With Multilevel Prediction:* As shown in Table II, FPN with multilevel prediction improves the AP from 59.45% to 60.07%. In particular, AP_{50} and AP_{75} are improved to 78.22% and 36.40%, respectively, which validate that the proposed component can locate ship objects more accurately than the baseline method. Meanwhile, both AP_S and AP_L are improved by nearly 3% at the cost of reducing AP_M from 67.66% to 63.57%. Thus, $P4$, $P3$, and $P2$ in our FPN effectively extract high-dimensional and high-resolution feature maps. Moreover, the combination of FPN and multilevel prediction helps to discriminate different scale targets and locate them with different head networks.
2) *Oriented Nonnormalized Gaussian Function:* To overcome the undesirable penalty for the negative location of the 2-D Gaussian function for oriented ship detection, an oriented nonnormalized Gaussian function is proposed. As shown in Table II, the proposed oriented Gaussian function improves AP from 60.07% to 60.43%. AP_{50}, in particular, and AP_{75} are improved to 78.55% and 36.92%, respectively. Thus, the introduction of the oriented Gaussian function helps us to locate the center point of ship objects with a high recall rate and increases the robustness of the center points to feature maps.
3) *Nonuniform Weighting for Loss Function:* Considering the imbalance distribution between different scale ship targets, nonuniform weighting with three weighting factors α_1, α_2, and α_3 for $P2$, $P3$, and $P4$ is devised. Both α_1 and α_2 are set as 1, while α_3 is set as 1.2 here. The detailed detection performance of these factors is analyzed as follows. One can see that the proposed nonuniform weighting strategy improves the AP from 60.43% to the highest value of 60.71%. In particular, AP_{50}, AP_{75}, AP_S, and AP_M achieved the highest value by combining our three key components, which fully demonstrate the effectiveness of our method.

What is more, inspired by the work in [21], the efficiency or complexity of these components is analyzed in this

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TABLE III
EXPERIMENTAL RESULTS OF DIFFERENT FPN STRUCTURE (%)

[0, m_2, m_3, m_4]	[0, 32, 64, ∞]	[0, 48, 96, ∞]	[0, 64, 128, ∞]
AP	60.18	**60.43**	60.38
AP_{50}	78.29	**78.55**	78.46
AP_{75}	36.50	**36.92**	36.69
AP_S	51.99	55.34	**56.47**
AP_M	65.14	**65.47**	65.41
AP_L	25.20	**30.85**	30.06

section. The efficiency is mainly determined by the model's parameters (Params) and calculations (multiadds), while the smaller the Params and multiadds, the higher the efficiency of the model. The efficiency comparisons of the baseline and our proposed components are shown in Table II. One can see that the baseline BBAV method has the Params of 53.42M and the multiadds of 305.19G. Introducing the FPN with multilevel prediction strategy into baseline, the Params is significantly reduced to 32.51M, while the multiadds is increased to 586.40G. The main reason lies in the fact that we replace the large-channel convolution kernels in baseline, e.g., 2048 channels, with more small-channel convolution kernels in FPN. Thus, the feature representation efficiency of the convolution layers in our method is significantly improved at the cost of increasing computation complexity. It is noticeable that the proposed oriented Gaussian function and nonuniform weight of loss function do not heavily increase Params and multiadds.

In the component of FPN with multilevel prediction, the scale factors $[0, m_2, m_3, m_4]$ assign different scale ships to different FPN head networks, e.g., $P2$, $P3$, and $P4$. To validate the performance of these factors, we choose three groups of parameter sets: $[0, 32, 64, \infty]$, $[0, 48, 96, \infty]$, and $[0, 64, 128, \infty]$. The corresponding results are shown in Table III. One can see that the factor $[0, 48, 96, \infty]$ achieves the highest AP, AP_{50}, AP_{75}, AP_M, and AP_L. The reason mainly lies in the fact that the factor set $[0, 32, 64, \infty]$ leads to an image scale that is too small in the feature maps, which severely affects the detection performance, and the factor set $[0, 64, 128, \infty]$ increases the imbalance between ship objects of different scales. Thus, the factors $[0, 48, 96, \infty]$ are used in our detector.

The weighting factors α_1, α_2, and α_3 in the nonuniform weighting component also played key roles. After a few trials, we choose four groups of weighting factors: [1, 1, 1], [1, 1.1, 1.2], [1, 1, 1.2], and [1, 1, 1.25]. This setting inspires fewer large samples than small samples and middle samples in HRSID, and we should increase the value α_3 to improve the effectiveness of large samples in the training process. The detection results of these parameters are shown in Table IV. One can see that the parameter [1, 1, 1.25] achieves the highest detection performance with AP values of 60.71% and AP_{50} 78.60%. Thus, we can set an appropriate weighting factor to suppress the imbalance between different scale samples and improve the detection performance.

C. Detection Results by Different Methods

To validate the effectiveness of the proposed method, we choose several typical and powerful DL-based oriented object detectors as the comparison methods. Specifically, these methods can be concluded as: 1) traditional DL methods, such as Faster RCNN [9] and RetinaNet for horizontal object detection (RetinaNet-H) [49]; 2) scene text detection methods such as R^2CNN [37]; 3) recently published methods fusing horizontal object detectors and oriented object detectors, including oriented-based RetinaNet (RetinaNet-O), oriented-based Faster RCNN (Faster RCNN-O), and ROI Transformer [30]; and 4) oriented detectors with a novel network structure and

TABLE IV
EXPERIMENTAL RESULTS OF DIFFERENT WEIGHTING FACTORS (%)

[α_1, α_2, α_3]	[1, 1, 1]	[1, 1.1, 1.2]	[1, 1, 1.2]	[1, 1, 1.25]
AP	60.43	60.16	60.34	**60.71**
AP_{50}	78.55	78.40	78.48	**78.60**
AP_{75}	36.92	36.54	36.85	**37.22**
AP_S	55.34	54.30	54.27	**56.86**
AP_M	65.47	66.49	65.07	**67.63**
AP_L	30.85	26.81	**32.48**	31.12

TABLE V
DETECTION ACCURACY OF DIFFERENT METHODS (%)

Method	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
RetinaNet-H	45.05	61.26	15.75	30.02	53.69	8.35
R^2CNN	50.17	69.72	22.59	26.68	47.71	9.96
R^3det	37.22	57.29	14.57	17.90	25.80	8.50
SCRDet	46.08	60.81	25.19	30.70	57.57	7.75
CSL	31.25	55.26	6.56	17.41	30.78	6.21
RetinaNet-O	52.71	68.96	28.33	9.09	32.77	11.51
Faster RCNN-O	**62.06**	77.39	34.87	32.32	42.76	10.58
ROI Transformer	60.65	76.82	**37.68**	37.04	48.94	17.14
BBAV	59.45	77.29	35.76	51.34	**67.66**	30.09
Proposed Method	60.71	**78.60**	37.22	**56.86**	67.63	**31.32**

TABLE VI
DETECTION EFFICIENCY OF DIFFERENT METHODS

Method	Params (M)	Multi-adds (G)
RetinaNet-H	32.33	161.75
R^2CNN	41.69	213.22
R^3det	40.67	203.43
SCRDet	43.07	298.87
CSL	41.04	205.29
RetinaNet-O	36.35	127.93
Faster RCNN-O	41.35	148.16
ROI Transformer	55.26	175.96
BBAV	53.42	305.19
Proposed Method	32.51	586.40

TABLE VII
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON SSDD+ (%)

Method	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
RetinaNet-H	16.41	36.04	3.03	8.61	7.99	1.70
R^2CNN	22.00	44.34	4.55	10.35	10.52	9.96
R^3det	16.41	36.04	3.03	8.61	7.99	1.70
SCRDet	16.68	36.50	1.52	9.11	12.57	1.47
CSL	15.49	33.36	2.60	7.73	6.21	2.09
DCL	19.50	42.36	4.55	8.93	7.86	2.07
RetinaNet-O	39.20	66.22	12.06	13.10	13.17	5.61
Faster RCNN-O	44.69	72.05	15.44	20.83	18.33	4.24
ROI Transformer	**46.58**	**72.72**	**16.91**	24.69	20.74	7.70
BBAV	37.33	61.72	11.49	18.68	28.72	9.21
Proposed Method	38.60	65.12	8.48	**33.01**	**31.65**	**26.31**

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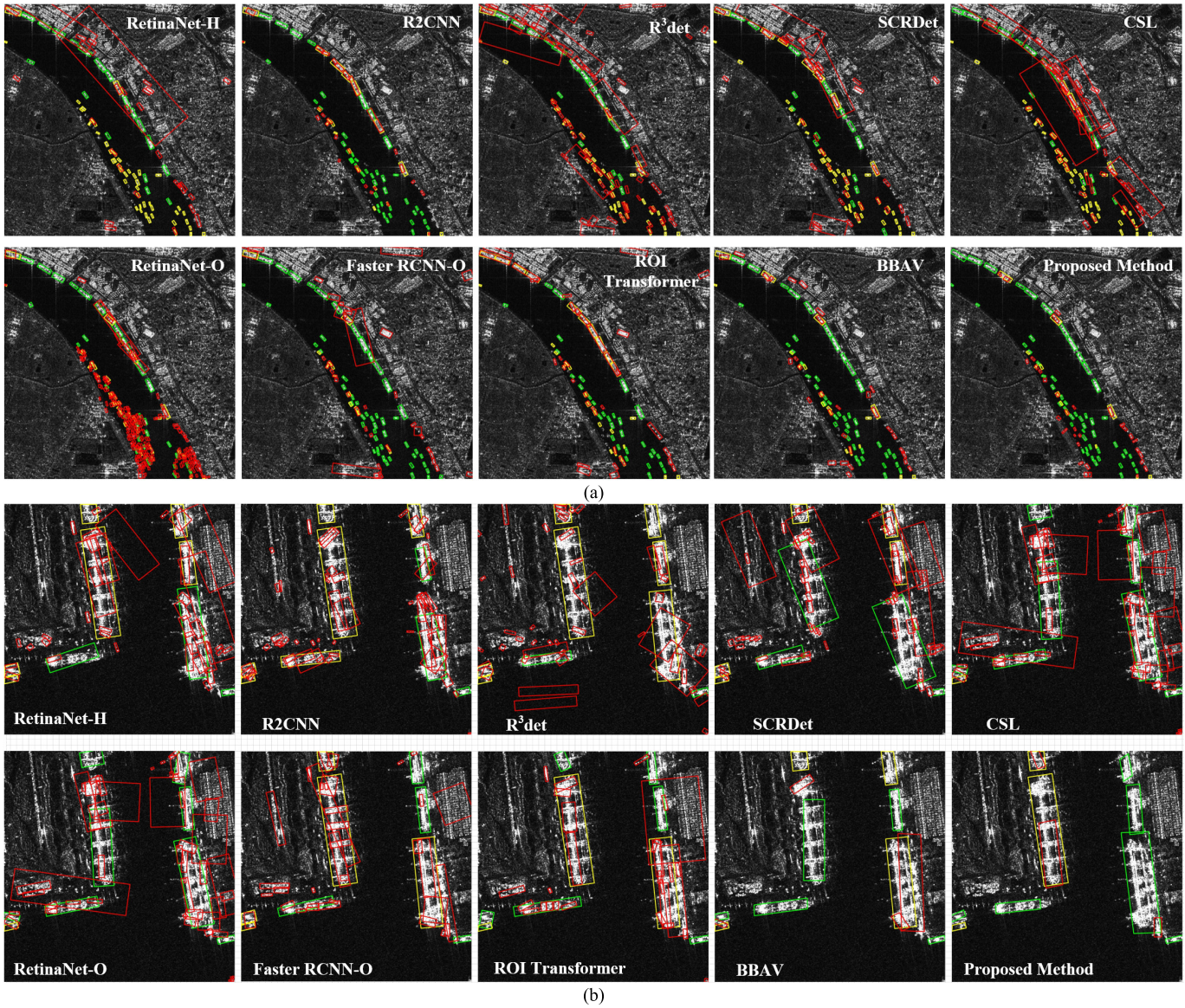


Fig. 13. Visualization comparison of different methods. The green boxes, red boxes, and yellow boxes indicate the correctly detected ships, false alarms, and the missing ships, respectively. (a) Detection results on dense distribution and inshore distribution. (b) Detection results of different scales ships.

optimization loss function, including R^3Det [36], detectors for Small, Cluttered and Rotated objects (SCRDet) [35], Circular Smooth Label (CSL) [40], and BBAV [41]. Among these methods, RetinaNet-H, R^2CNN, SCRDet, R^3Det, and CSL are executed by the published TensorFlow-based rotation detection benchmark UranusDet. Except for the class number and data settings, all the network settings and parameter settings for our oriented ship detection of these methods remain the same as those in UranusDet. RetinaNet-O, Faster RCNN-O, and ROI Transformer are executed by the aerial detection framework based on PyTorch. Their parameters and network structure also remain the same as those in AerialDetection. Moreover, all these detectors take ResNet-50 as the backbone. The detection accuracy of these methods is shown in Table V.

In Table V, RetinaNet-H achieves 61.26% AP_{50}, which demonstrates that this detector has good performance for oriented ship detection. Regarding AP_{75}, the accuracy is only 15.75%; thus, the horizontal object detector cannot be directly used for high-accuracy oriented ship detection. The scene text detection method R^2CNN also achieves high accuracy with an AP of 50.17% and AP_{50} of 69.72%. Different from the above methods, the detectors devised for optical object detection, such as SCRDet, R^3Det, and DCL, do not achieve preferable detection performance because the highest AP is 46.08%. The reason mainly lies in the large characteristic difference between optical images and SAR images. Meanwhile, the speckle noise in SAR images also affects the detection performance. Thus, these optical-devised detectors are not suitable for oriented ship detection in SAR images. Notably, methods fusing horizontal object detectors and oriented object detectors, such as RetinaNet-O, Faster RCNN-O, and ROI Transformer, have achieved higher accuracy than other methods. Specifically, Faster RCNN-O achieves the highest AP of 62.06%, while ROI Transformer achieves the highest AP_{75}

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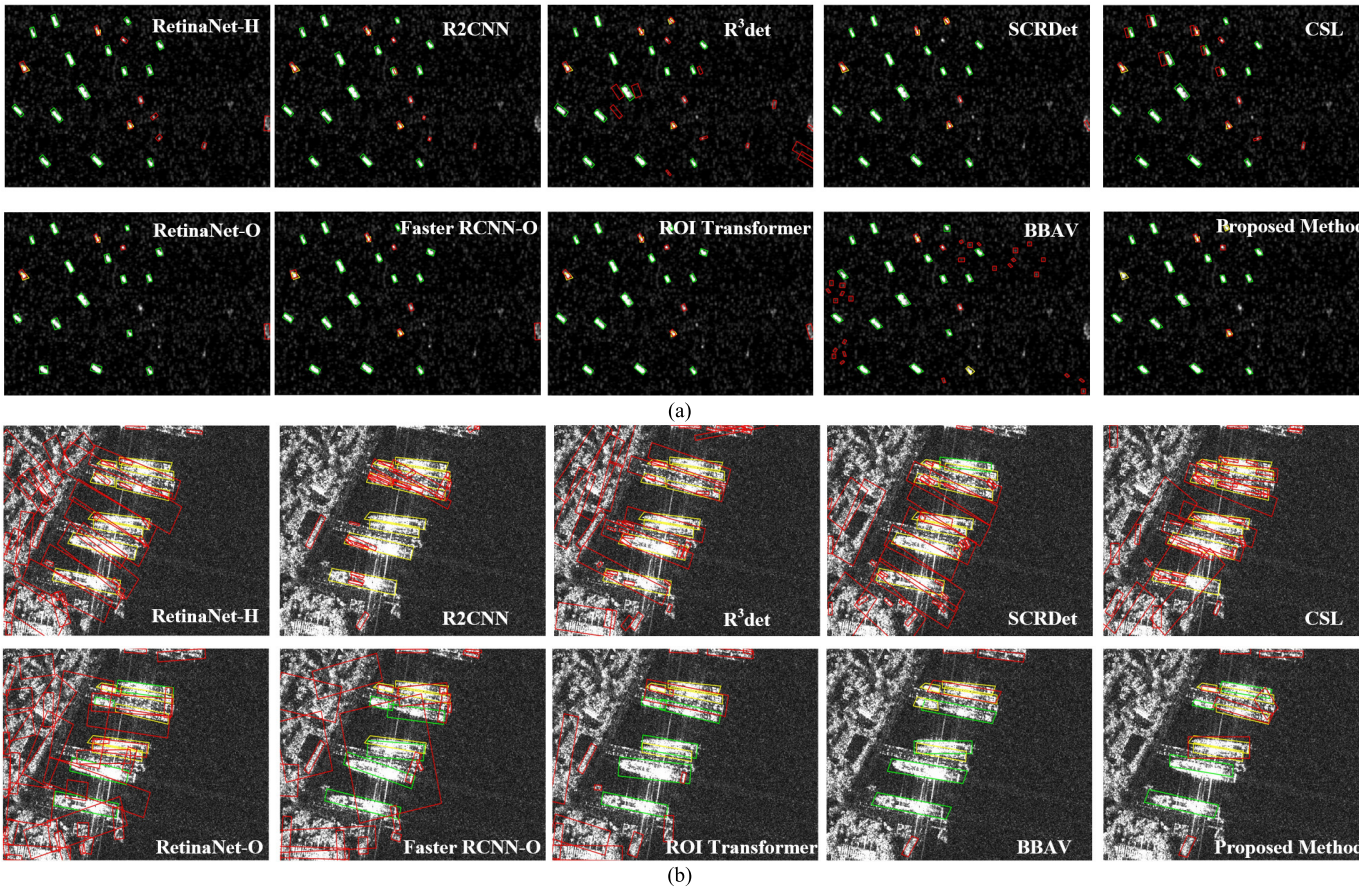


Fig. 14. Visualization comparison of different methods on two image in SSDD+. (a) Results on dense distribution. (b) Results on inshore region.

of 37.68%. In addition to the above methods, the center point-based BBAV also achieves satisfactory detection results with an AP_{50} of 77.29% and an AP_M of 67.66%, which demonstrates the feasibility of locating oriented ships in SAR images by detecting the center points. Inspired by this strategy, we propose a novel detector, and the detection result is shown in Table V. The proposed method achieves the highest AP_{50} of 78.60% and very high AP and AP_{75}. Our method also achieves the highest AP_S of 56.86% and AP_L of 31.32%. This performance improvement is mainly due to our contributions, such as the combination of FPN and multi prediction, the oriented non-normalized Gaussian function, and the nonuniform weighting of different level loss functions.

The detection efficiency of these methods is shown in Table VI. The proposed method has nearly the lowest Params. The reason mainly lies in the fact that the proposed anchor-free detectors do not devise complicated anchors and only use simplified convolution kernels in the detection network. It is noticeable that these kernels also lead to the highest multiadds of our detector than other methods. According to the above analysis, the proposed detector method has better feature representation efficiency and a slower running speed than other methods.

In addition to the above quantitative results in Table V, we also choose two SAR images from HRSID testing data to understand the detection of different methods. The visualization results with IOU 0.5 are shown in Fig. 13, where Fig. 13(a) shows the detection results on dense distribution and inshore distribution, and Fig. 13(b) shows the detection results of ships of different scales. In Fig. 13(a), RetinaNet-H and R^2CNN can detect some ships moored on the shore and ships on the water body. R^3Det, SCRDet, CSL, and RetinaNet can detect some salient ships. However, the false alarm rate of these methods is too high, which severely affects the detection performance. Compared with the above methods, the Faster RCNN-O, ROI Transformer, and BBAV can correctly detect more ships while keeping a low false alarm rate. In particular, the proposed method detects the most ships and generates the fewest false alarm ships. We also choose the SAR image in Fig. 13(b) to validate the detection results on ships of different scales. One can see that the RetinaNet-H, R^2CNN, R^3Det, Faster RCNN-O, and ROI Transformer cannot locate ships with different scales. Meanwhile, although SCRDet, CSL, and RetinaNet-O can detect large, middle, and small ships, the false alarm caused by land background is too high. Compared with these methods, BBAV and our method have higher detection performance. In particular, the proposed method has the lowest false alarm and the highest detection rate. According to the above analysis, the proposed method has good detection performance for dense distribution, inshore distribution, and ships with different scales. It is noticeable that there are also some false

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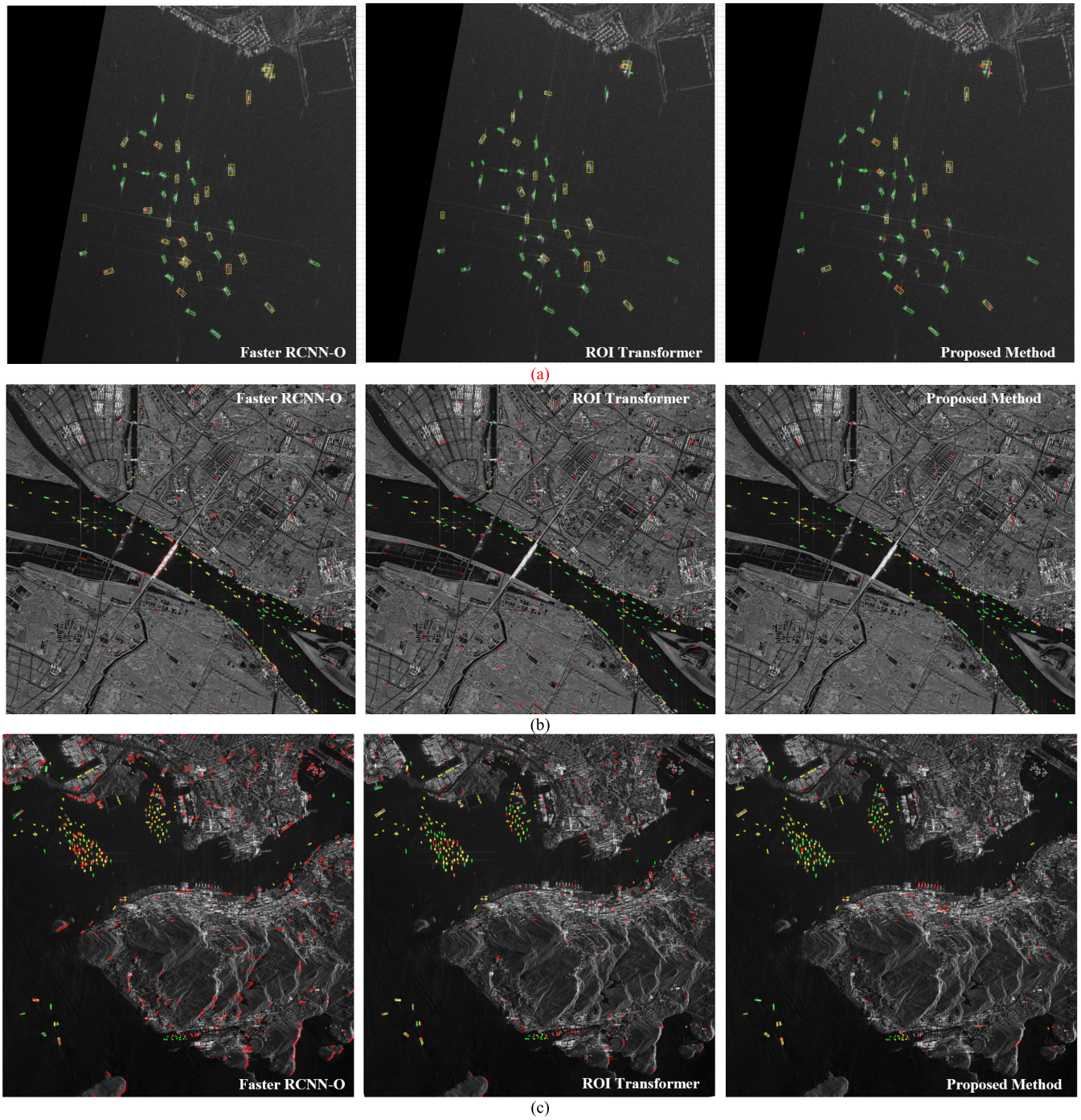


Fig. 15. Visualization comparison of different methods on three large-scene images. (a) Results on an AIR-SARShip-1.0 image. (b) Results on the Yangtze River region in Nanjing city of China. (c) Results on coastal region in Hong Kong city of China.

alarm targets and missing alarm targets in the results of our methods, such as too small ships in Fig. 13(a) and inshore ships in Fig. 13(b). These undesirable results encourage us to further improve the detection performance of our method in the future.

D. Extended Experimental Results on SSDD+

In Section IV-C, both quantitative results and visualization results on the testing data of HRSID demonstrate the superiority of the proposed method over other detection methods. In addition to these results, we introduce an extended experiment on SSDD+ to explore the detection robustness of the different methods. SSDD+ is a modified version of SSDD with OBB annotation. Specifically, different from traditional SAR ship detection methods where training samples and testing samples come from the same dataset, we use the well-trained detection model of different detectors on HRSID training data to infer all 1160 SAR images in SSDD+. Since that, the images in HRSID and SSDD+ come from different SAR systems, and the scale distribution and resolution distribution

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between them also differ from each other. Thus, this proposed extended experiment can effectively validate the detection robustness of different methods.

Using the well-trained model to test the SSDD+ images, the corresponding results are shown in Table VII. One can see that all the AP of different methods has been decreased than that in Table VI. The reason mainly lies in the fact that there exists a significant difference between HRSID images and SSDD+ images. RetinaNet-H, R^2CNN, R^3Det, SCRDet, CSL, and DCL do not achieve satisfactory detection results. In particular, Faster RCNN-O achieves the highest AP of 44.68%, AP_{50} of 72.72%, and AP_{75} of 16.91%. Although the proposed method does not achieve the highest values of AP, AP_{50}, and AP_{75}, the values of AP_S, AP_M, and AP_L achieved the highest values by the proposed method. Thus, the proposed methods can effectively discriminate ships of different scales than other methods. To intuitively understand the detection performance on SSDD+, we choose two images from these data, which include the dense distribution and the inshore region. In Fig. 14(a), there are lots of ships on a small scale. One can see that the proposed method can detect most ships while keeping a low false alarm rate. The inshore region in Fig. 14(b) also demonstrates that the proposed method can locate the ships at the inshore and harbor regions. Thus, the proposed method has good robustness to different SAR ship detection data and multiresolution SAR images. Some densely distributed ships are missed in our detection results which may be caused by different data distribution and ineffective feature extraction for these ships.

E. Extended Experimental Results on Large-Scale Images

To further understand the detection result on large-scene SAR images, we choose three large-scene SAR images, where one from the AIR-SARShip-1.0 data, as shown in Fig. 15(a), and the other two images from the Gaofen-3 system, as shown in Fig. 15(b) and (c). Fig. 15(a) with the size of 3000 × 3000 pixels is located in a harbor and includes different scale ships. Fig. 15(b) and (c) images with the resolution of 1 m × 1 m were acquired under spotlight mode and HH polarization. Among them, Fig. 15(b) was acquired on the Yangtze River in Nanjing city, while some ships are anchored on the shore and some ships are driving in this river. Fig. 15(c) was acquired in Hong Kong, while some ships are densely anchored on harbors and some ships are driving in the sea. The Faster RCNN-O and ROI Transformer are taken as the comparison methods for their good robustness in SSDD+. Using the well-trained model on HRSID to test these images, the detection results of these methods are shown in Fig. 15. In the testing process, we first crop these big images into image patches of size 800 × 800 pixels with the overlap 200 pixels, then the well-trained models are used to detect ships in these patches, and finally, the detection results on patches are combined into results on big images as the cropping order. The NMS processing is used to remove the repetitive ships. In Fig. 15(a), although all three methods do not use the images in AIR-SARShip data for training, they can still determine the position and oriented angle of ships. Among them, the proposed method accurately detects more ships than Faster RCNN-O and ROI Transformer method while keeping a low false alarm. In Fig. 15(b), although Faster RCNN-O and ROI Transformer detect most ships on the shore and river, too many ships are wrongly detected in the land region. Compared with these methods, the proposed method detects more ships and leads to lower false alarms. In Fig. 15(c), Faster RCNN-O generates too many false ships in complex land regions, while the proposed method detects most ships and leads to the lowest false alarm.

TABLE VIII
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON GAOFEN-3 IMAGES

Method	Recall Rate (%)	Precision (%)	F1 score (%)
Faster RCNN-O	17.46	33.64	22.99
ROI transformer	24.16	41.56	30.56
Proposed method	31.10	43.33	36.21

The quantitation results corresponding to Fig. 15 are shown in Table VIII. The proposed method achieves the highest recall rate and precision. In particular, the $F1$ score of the proposed method is 36.21%, which is higher than that of the Faster RCNN-O and ROI Transformer. Thus, both the visualization results and quantitation results demonstrate the robustness of the proposed method.

V. CONCLUSION

In this article, we propose the oriented Gaussian function-based BBAVs for oriented ship detection in multiresolution SAR images. The proposed method firstly extracts multidimensional features from SAR images with a backbone network and FPN. Then, we detect ships of different scales on different levels of feature maps with a head network structure that consists of a classification network and a regression network, where the classification subnetwork determines the center pixel of ships and the regression subnetwork regresses the oriented bounding box for each ship. Compared with other methods, there are three key modules of our method: the combination of FPN and multilevel prediction for detecting multiscale ships, the oriented nonnormalized Gaussian function for locating the center point of ship targets, and the nonuniform weighting of the loss function for suppressing the imbalanced sample distribution. Experimental results on HRSID fully demonstrate the effectiveness and efficiency of our contributions. Moreover, the extended experiments on SSDD+, AIR-SARShip-1.0, and Gaofen-3 images also validate that our method has good robustness to different SAR systems and different resolutions. In the future, we will continue this research to improve the performance of our detection method under various and special conditions, such as densely distributed ships, inshore ships, and complicated backgrounds.

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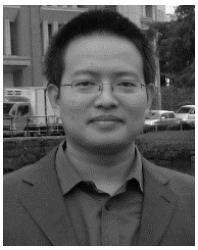
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